

Matrix product states: the Schmidt decomposition at every bond

FYS4480/FYS9480 – Week 40, October 1–2, 2026

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Thursday October 1: ten minutes of repetition – Thouless' theorem and the stability matrix; then entanglement across a cut, area laws, and the matrix product state from $L - 1$ singular value decompositions

Friday October 2: canonical forms, the transfer matrix and the cost of contractions; matrix product operators as finite-state automata; adding, applying and compressing; fermions on a chain

Reading: these slides; lecture notes chapter 11, sections 11.1–11.4 and 11.7 (chapter 6, sections 6.7 and 6.10 for the repetition); chapter 1, sections 1.28–1.31. Code: `BookPrograms/chapterdmrg/dmrg.py`

Plan for week 40

Thursday October 1, 2×45 min: from entanglement to matrix product states

- Repetition (10 min): Thouless' theorem, the energy to second order, the stability matrix $\hat{M}$ and its blocks Δ , A , B ; Lipkin unstable, pairing stable and useless (sections 6.7, 6.10–6.11)
- A third strategy: restrict the *entanglement*, not the basis; the methods in one table (chapter 11, introduction)
- A bipartite state is a matrix; Schmidt decomposition = SVD; isometries; determinants are entangled (section 11.1)
- Area laws: $S \sim \ell^{D-1}$, what is a theorem, the logarithm at criticality, which states are compressible (section 11.1)
- The MPS from $L - 1$ SVDs; the bond index; GHZ, W and AKLT by hand; truncation and the discarded weight (section 11.2)

Friday October 2, 2×45 min: MPS technology and operators

- Gauge freedom and canonical forms; why the block states are orthonormal; norms, expectation values and the transfer matrix; the cost is in the order (section 11.2)
- Matrix product operators as finite-state automata: Heisenberg $D = 5$, pairing $D = 4$, the systematic construction (section 11.3); adding, applying, compressing (section 11.4)
- Fermions on a chain: Jordan–Wigner, free in D , expensive in locality (section 11.7)
- Exercises: suggested set for week 40 (the stability matrix by Wick's theorem, positive semidefinite matrices) and answers to the week-39 set (last frames)

Repetition: Thouless' theorem and stability

Where week 39 ended: one condition, three derivations (chapter 6)

The Hartree-Fock condition

$$\langle a|\hat{f}|i\rangle = \langle a|\hat{h}_0|i\rangle + \sum_{j \leq F} \langle aj|\hat{v}|ij\rangle_{AS} = 0 \quad \text{for all } i \leq F, a > F,$$

equivalently $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = f_{ia} = 0$ (Brillouin). Derived three times: coefficients with multipliers (6.4), one orbital $\phi_i \rightarrow \phi_i + \eta\phi_a$ (6.6), the determinant as a whole (6.7).

Thouless' theorem (6.7)

Every determinant with $\langle c|c'\rangle \neq 0$ is

$$|c'\rangle = e^{\hat{T}}|c\rangle, \quad \hat{T} = \sum_{a>F} \sum_{i \leq F} C_{ai} a_a^\dagger a_i,$$

$(n - N)N$ complex parameters and nothing else;
 $e^{\hat{T}} = \prod_i (1 + \hat{T}_i)$, and the cross terms $\hat{T}_i \hat{T}_j$ are the $2p$ - $2h$ admixtures, second order in C .

- First order in δC : $f_{ai} = 0$ again – the singles exhaust the neighbourhood of a determinant.
- Second order: does the energy *rise* in every direction? Stationary is not minimal.
- That second-order question is what we repeat now, because its answer – a matrix – is what the exercises of this week compute with Wick's theorem.

Many-body connection

Today's chapter (11) changes the question. Hartree-Fock asks "which determinant"; every method after it asks "what to do with the $2p$ - $2h$ block". The DMRG asks neither: it has no reference at all.

The energy to second order, and the stability matrix (6.10)

With $|c'\rangle = e^{\sum \delta C_{ai} a_a^\dagger a_i} |c\rangle$, intermediate normalisation and $f_{ai} = 0$,

$$\langle c'|c'\rangle = 1 + \sum_{ai} |\delta C_{ai}|^2 + \mathcal{O}(\delta C^3), \quad \frac{\langle c'|\hat{H}|c'\rangle}{\langle c'|c'\rangle} = E_0 + \frac{\Delta E}{1 + \sum_{ai} |\delta C_{ai}|^2}, \quad \Delta E = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle.$$

Three Wick calculations, three blocks

$$\begin{aligned} \langle c | a_i^\dagger a_a \hat{H} a_b^\dagger a_j | c \rangle - E_0 \delta_{ab} \delta_{ij} &= (\Delta + A)_{ai,bj}, & \langle c | \hat{H} a_a^\dagger a_i a_b^\dagger a_j | c \rangle &= B_{ai,bj}^*, \\ \Delta_{ai,bj} &= (\varepsilon_a - \varepsilon_i) \delta_{ab} \delta_{ij}, & A_{ai,bj} &= -\langle aj | \hat{v} | bi \rangle_{AS} = \langle aj | \hat{v} | ib \rangle_{AS}, & B_{ai,bj} &= \langle ab | \hat{v} | ij \rangle_{AS}, \\ \chi &= \begin{bmatrix} \delta C \\ \delta C^* \end{bmatrix}, & \hat{M} &= \begin{pmatrix} \Delta + A & B \\ B^* & \Delta + A^* \end{pmatrix}. \end{aligned}$$

- χ carries δC and δC^* because the quadratic form has $\delta C^* \delta C$ terms (A) and $\delta C \delta C$, $\delta C^* \delta C^*$ terms (B): a real quadratic form in $\text{Re } \delta C$, $\text{Im } \delta C$ written in complex variables.
- $\hat{M}$ is Hermitian, and the denominator is positive, so the sign of ΔE is the sign of $\langle \chi | \hat{M} | \chi \rangle$.

The criterion, what is necessary, and what it means

Stability

The Hartree-Fock determinant is a local minimum on the manifold of determinants **if and only if** $\hat{M} \geq 0$, i.e. every eigenvalue of $\hat{M}$ is non-negative – $\hat{M}$ is positive semidefinite.

- **Necessary, not sufficient:** restrict χ to its upper half: $\Delta + A \geq 0$. This is the Hamiltonian in the $1p\text{-}1h$ space measured from E_0 – the TDA/CIS matrix. Cheap, and the usual first check.
- **Even cheaper, even weaker:** the diagonal, $\varepsilon_a - \varepsilon_i + A_{ai,ai} = \langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle - E_0 \geq 0$: every single excitation must cost energy. (Week 39, Thursday: the curvature of $E(\eta)$ along one direction.)
- **Sufficient only with B :** the B block couples δC to δC^* , i.e. two excitations at once; it can pull an eigenvalue of $\hat{M}$ below zero while $\Delta + A > 0$ (Lipkin).

Many-body connection

$\hat{M}$ is the RPA matrix of chapter 7: “the mean field is stable” = “all RPA frequencies are real”. A negative eigenvalue is not a numerical accident; its eigenvector χ is the *direction* in which to deform the reference, usually by breaking a symmetry of $\hat{H}$.

A reminder on positive semidefinite matrices, and the necessary conditions one can read off without diagonalising, is in exercise 2 of this week.

Two examples, and the bridge to this week

Lipkin, $W = 0$ (6.11): unstable beyond $\chi = 1$

Interaction changes J_z by two: $A = 0$, $\Delta = \varepsilon_1$, $B = V(J - 1)$ with largest eigenvalue $|V|(N - 1)$, so

$$\lambda_{\min}(\hat{M}) = \varepsilon - |V|(N - 1) = \varepsilon(1 - \chi).$$

$\Delta + A > 0$ always – the necessary condition never fires – yet the spherical solution is a saddle for $\chi > 1$. Energy flat at $-\varepsilon/2$ per particle; only the curvature notices.

- Lesson: satisfying $f_{ai} = 0$ is not enough; the second derivative decides; symmetry breaking is a device whose bill comes later.
- **This week** we leave the reference behind: the DMRG restricts the *entanglement* of the state, not the excitation level about a determinant. The pairing model will return in section 11.8, where the DMRG finds the seniority-zero ground state without being told about pairs.

Pairing, four levels, four pairs: stable and useless

$A = 0$, $\Delta = \varepsilon_q - \varepsilon_p + g/2$, $B = -g/2$ between $\delta C_{q\uparrow p\uparrow}$ and $\delta C_{q\downarrow p\downarrow}^*$: 2×2 blocks with eigenvalues $\varepsilon_q - \varepsilon_p$ and $\varepsilon_q - \varepsilon_p + g$,

$$\lambda_{\min}(\hat{M}) = \varepsilon_3 - \varepsilon_2 = 1 \quad \text{for every } g.$$

No number-conserving mean field sees pair correlations; the correlation is in the doubles.

Thursday, lecture 1: Entanglement, area laws and the Schmidt decomposition

Where we are (chapter 11, introduction)

Chapters 5–6: shrink the basis

- FCI: all $\binom{n}{N}$ determinants
- CISD: up to a fixed excitation level
- HF: one determinant, best orbitals

All say: *most configurations do not matter.*

This week: a property of the *state*

- Local + gapped + 1D $\Rightarrow$ bounded entanglement across a cut
- Bounded entanglement $\Rightarrow$ few parameters
- Tool: the SVD, at **every bond**

L sites, local basis $|s_i\rangle$ with $s_i = 1, \dots, d$; χ : bond dimension of a state, D : of an operator.

No reference determinant appears anywhere.

The methods you have, in one table (chapter 11, Table 11.1)

	ansatz	varied	size	size consistent
HF	$ \Phi_0\rangle$	orbitals	$n \times n$ eigenproblem	yes
CIS	$(1 + \hat{C}_1) \Phi_0\rangle$	coefficients	$1 + N(n - N)$	—
CID	$(1 + \hat{C}_2) \Phi_0\rangle$	coefficients	$\mathcal{O}(n^6)$	no
CISD	$(1 + \hat{C}_1 + \hat{C}_2) \Phi_0\rangle$	coefficients	$\mathcal{O}(n^6)$	no
FCI	$(1 + \dots + \hat{C}_N) \Phi_0\rangle$	coefficients	$\binom{n}{N}$	yes
DMRG	matrix product state	tensors	$Ld\chi^2$ parameters	yes

- CI rows are controlled by an *excitation level*: a statement about a reference
- The DMRG row is controlled by *entanglement*: a statement about the state

A bipartite state *is* a matrix (section 11.1; chapter 1, section 1.30)

$$|\Psi\rangle = \sum_{i,j} C_{ij} |i\rangle_A \otimes |j\rangle_B, \quad C = U \Sigma V^\dagger$$

Schmidt decomposition

$$|\Psi\rangle = \sum_{k=1}^r \sigma_k |u_k\rangle_A \otimes |v_k\rangle_B, \quad |u_k\rangle_A = \sum_i U_{ik} |i\rangle_A, \quad |v_k\rangle_B = \sum_j V_{jk}^* |j\rangle_B$$
$$\rho_A = C C^\dagger, \quad \rho_B = C^\dagger C, \quad S_A = - \sum_k \sigma_k^2 \ln \sigma_k^2 = S_B$$

- r : the Schmidt rank. Both ρ 's have the same eigenvalues σ_k^2
- **Mind the conjugate:** $C_{ij} = \sum_k U_{ik} \sigma_k V_{jk}^*$ – invisible for real C , an $\mathcal{O}(1)$ error otherwise

What U and V are (section 11.1)

Thin SVD, rank r : U is $d_A \times r$, V is $d_B \times r$.

$$U^\dagger U = I_r = V^\dagger V \quad \text{but} \quad UU^\dagger = P_{\text{col}}, \quad VV^\dagger = P_{\text{row}}$$

- Not square, so **not unitary**: they are *isometries*
- $U^\dagger U = I$ is exactly $\langle u_k | u_{k'} \rangle = \delta_{kk'}$
- Tomorrow the same equation is called *left-canonical*
- A tensors come from U , B tensors from $V^\dagger$

Three readings

Structural: $r = 1 \iff$ product state $\iff S_A = 0$ *Numerical*: the χ largest σ_k give the best rank- χ approximation *Physical*: $S_A \leq \ln \min(d_A, d_B)$

Pause to think: rank, products and determinants

Pause to think #1

(about 4 min – discuss with your neighbour)

1. Show that Schmidt rank $r = 1$ means $|\Psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B$. What does it say about C ?
2. A Hartree product has $S = 0$ across every cut. Is the same true of a *Slater determinant*? Cut between orbitals 2 and 3 in

$$a_1^\dagger a_2^\dagger |0\rangle \quad \text{and} \quad \frac{1}{\sqrt{2}} (a_1^\dagger a_2^\dagger + a_3^\dagger a_4^\dagger) |0\rangle.$$

Answer to pause #1

1. $C = \sigma_1 u_1 v_1^\dagger$: a *rank-one* matrix, $C_{ij} = u_i v_j^*$. Entanglement is rank greater than one.
2. It depends on the cut:

$$a_1^\dagger a_2^\dagger |0\rangle = |11\rangle_A |00\rangle_B : \quad r = 1, \quad S = 0$$

$$\frac{1}{\sqrt{2}} (|11\rangle_A |00\rangle_B + |00\rangle_A |11\rangle_B) : \quad r = 2, \quad S = \ln 2$$

A generic determinant needs rank up to $2^{\min(N, L-N)}$. “Uncorrelated” (ch. 2) $\neq$ “unentangled”.

Five states to keep in your head (section 11.1)

State	Schmidt coefficients	rank	S_A
$ 0\rangle \otimes 1\rangle$	$\{1\}$	1	0
$(00\rangle + 11\rangle)/\sqrt{2}$	$\{1/\sqrt{2}, 1/\sqrt{2}\}$	2	$\ln 2 = 0.693$
$\cos\theta 00\rangle + \sin\theta 11\rangle$	$\{\cos\theta, \sin\theta\}$	2	$0 \rightarrow \ln 2$
Heisenberg, $L = 10$, middle cut	0.497, 0.497, 0.0029, ...	32	0.738
Pairing, 12 levels, middle cut	0.517, 0.409, ...	64	0.924

- Last row is the pairing model of chapter 1 (section 1.30): ten Schmidt states capture all but 8×10^{-7} , out of a possible $\ln 64 = 4.16$; the Heisenberg row is the worked example of section 11.1
- The question is not *how many determinants*, but **how fast the Schmidt spectrum decays**

Area laws: the entropy scales with the *boundary* (section 11.1)

Block A of linear size ℓ in $\mathcal{D}$ spatial dimensions:

generic state: $S_A \sim \ell^{\mathcal{D}}$ (**volume**) gapped local ground state: $S_A \sim \ell^{\mathcal{D}-1}$ (**boundary**)

$\mathcal{D}$	block	boundary	S (gapped)	bond dimension needed
1	ℓ sites	2 points	constant	constant, at any L
2	$\ell \times \ell$	4ℓ	$\sim \ell$	$\chi \sim e^{\alpha\ell}$
3	ℓ^3	$6\ell^2$	$\sim \ell^2$	$\chi \sim e^{\alpha\ell^2}$

- $\chi \sim e^S$: the area law **is** the statement that the ansatz is affordable – and only $\mathcal{D} = 1$ makes it L -independent
- $\mathcal{D} = 1$, gapped: a theorem (Hastings). $\mathcal{D} \geq 2$: expected, proven only in special cases
- ($\mathcal{D}$ is the spatial dimension; D stays reserved for the operator bond dimension of Friday)

Corrections to the area law, and what it does *not* say (section 11.1)

Logarithms: where the law is violated mildly

Critical chain ($\mathcal{D} = 1$, gapless), block of ℓ sites in an open chain of L :

$$S_\ell \simeq \frac{c}{6} \ln \left[\frac{2L}{\pi} \sin \frac{\pi \ell}{L} \right] + s_0,$$

with c the central charge. Then χ grows as a *power* of L – slow enough that the method still works. Gapless fermions in $\mathcal{D}$ dimensions keep the logarithm: $S \sim \ell^{\mathcal{D}-1} \ln \ell$.

A trap to avoid

“Entanglement is limited by locality” is **not** a theorem – the gap and $\mathcal{D} = 1$ do the work. *Excited* states of local Hamiltonians obey a volume law in any dimension; Motzkin and Fredkin chains reach $\sqrt{L}$; and the pairing and Lipkin models, which the DMRG handles well, are long range – they work by *symmetry and factorisation*, not locality.

Pause to think: two states, one cut

Pause to think #2

(about 4 min – discuss with your neighbour)

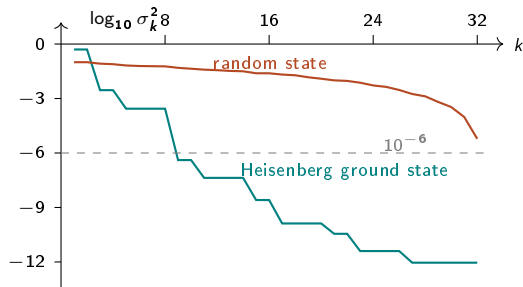
Two normalised states of $L = 10$ qubits: a random vector, and the ground state of the open Heisenberg chain. Cut in the middle, so C is 32×32 .

1. What is the largest entropy either could have?
2. Guess S for each, and how many Schmidt values you must keep to lose less than 10^{-6} of the norm.
3. If one needs all 32 – what does that say about compressing it by *any* method?

Answer: which state is compressible?

state	S	χ for $\varepsilon < 10^{-6}$
random	≈ 2.95	32 (all)
Heisenberg	0.7379	8

- Maximum: $\ln 32 = 3.4657$
- Random: **incompressible**
- Ground state: two weights of 0.497, then a drop of two decades

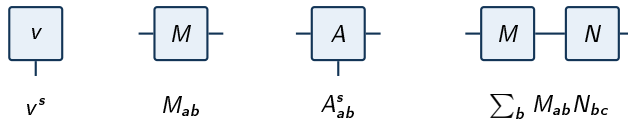


Exponential decay is what makes the method work.

Thursday, lecture 1 continued: The MPS, its bond dimension and its truncation

Tensor-network diagrams (section 11.2)

A shape is a tensor, a leg is an index, a *connected* leg is a summed index.



- A closed diagram is a number; k free legs is a k -index tensor
- Contracting two tensors costs the product of all their dimensions
- The **order** of contractions decides the cost: χ^3 or χ^4
- Every SVD below is preceded by a *reshape*

Building the MPS: SVD, $L - 1$ times (section 11.2)

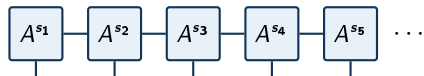
$|\Psi\rangle = \sum_{s_1 \dots s_L} C_{s_1 \dots s_L} |s_1 \dots s_L\rangle$: a tensor with L legs, d^L entries.

$$C_{s_1, (s_2 \dots s_L)} = \sum_{a_1} \underbrace{U_{s_1 a_1}}_{A_{a_1}^{s_1}} \sigma_{a_1} V_{a_1, (s_2 \dots s_L)}^\dagger \quad \longrightarrow \quad \text{reshape, SVD again, continue}$$

After $L - 1$ steps

$$C_{s_1 s_2 \dots s_L} = A^{s_1} A^{s_2} \dots A^{s_L}$$

L matrices, one per site, *selected* by the local index s_i ; bond dimension 1 at the ends.



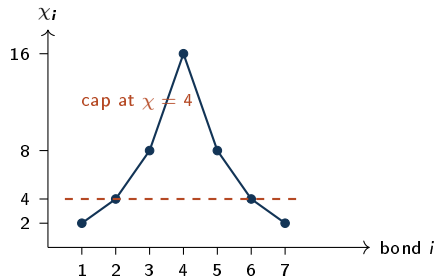
Nothing has been approximated yet – the construction is exact for any state.

What the bond index is (section 11.2)

A^{s_i} has shape $\chi_{i-1} \times \chi_i$, with

$$\chi_i = \min(d^i, d^{L-i}, \chi).$$

- The exact Schmidt rank of the cut at bond i
- Three arguments, three limits: left half, right half, our choice
- The profile is a tent, peaking at $d^{L/2}$
- So $\chi_{i-1} \times \chi_i$ is right on *both* halves



$L = 8$ qubits: 2, 4, 8, 16, 8, 4, 2. A cap at $\chi = 4$ touches only the three middle bonds.

MPS by hand: GHZ and W (section 11.2; chapter 11, exercise 1)

GHZ ($|000\rangle + |111\rangle$)/ $\sqrt{2}$ – the bond index as a classical memory

$$A_1^0 = \frac{1}{\sqrt{2}}(1, 0), \quad A_1^1 = \frac{1}{\sqrt{2}}(0, 1), \quad A_2^0 = \text{diag}(1, 0), \quad A_2^1 = \text{diag}(0, 1), \quad A_3^0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A_3^1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Diagonal in the middle: the branch may not change. $\chi = 2$, and no $\chi = 1$ MPS can do it.

W ($|100\rangle + |010\rangle + |001\rangle$)/ $\sqrt{3}$ – the bond index as an automaton

$$A_2^0 = I, \quad A_2^1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \implies C_{110} = 0$$

State 1 = “not placed yet”, state 2 = “done”. **Remember this** – tomorrow it builds every operator.

$\chi = 1$ is a product state; every bond dimension above one is entanglement.

A physical state that is exactly an MPS: AKLT (section 11.2; chapter 11, exercise 3)

$$\hat{H} = \sum_i \left[S_i \cdot S_{i+1} + \frac{1}{3} (S_i \cdot S_{i+1})^2 \right]$$

$$A^{+1} = \sqrt{\frac{2}{3}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad A^0 = -\sqrt{\frac{1}{3}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad A^{-1} = -\sqrt{\frac{2}{3}} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Spin-1, $d = 3$, $\chi = 2$, and $\sum_s A^{s\dagger} A^s = I$: left-canonical as they stand.

Checked in a few lines of numpy (chapter 11, exercise 3d)

- $\chi = 2$ is *exact* at every L : $\hat{H}|\psi\rangle = E|\psi\rangle$ to 10^{-15} , $E = -\frac{2}{3}(L-1)$ on the open chain ($-\frac{2}{3}L$ on a ring): $-\frac{2}{3}$ per bond
- $S = \ln 2$ on an open chain, $2 \ln 2$ on a ring – the entropy counts boundaries
- Transfer matrix eigenvalues 1 and $-\frac{1}{3}$: $\xi = 1/\ln 3 = 0.9102$

Pause to think: bond dimension

Pause to think #3

(about 4 min – discuss with your neighbour)

1. GHZ and AKLT both have $\chi = 2$ at any L , so both are cheap. Does that mean they are weakly entangled? What exactly is bounded?
2. An MPS has $\approx Ld\chi^2$ parameters, the exact one needs $\chi = d^{L/2}$. For $L = 12$, $d = 4$: at what χ does the MPS stop being a saving?

Answer to pause #3

1. Bounded is the entropy *across a single cut*, $\ln 2$ for both. GHZ superposes two macroscopic branches; AKLT has string order and spin- $\frac{1}{2}$ edge modes. Cheap means few Schmidt states per cut, not “nearly a product state”.
2. $\chi > \sqrt{4^{12}/48} = 591$, while the exact MPS needs $4^6 = 4096$: an *exact* MPS costs more than storing the tensor. The ansatz pays off only because real spectra decay.

Counting parameters, and truncation (section 11.2)

$Ld\chi^2$ parameters, **linear in L** against d^L .

The error you control

Keep the χ largest singular values at every bond: the best rank- χ approximation (Eckart–Young), with discarded weight

$$\varepsilon_\chi = \sum_{k>\chi} \sigma_k^2.$$

- Report ε_χ , not χ
- *Variational*: $E(\chi) \geq E_0$, improving monotonically with χ
- **The method computes its own error bar** – an excitation level does not

Numerical check: truncating the exact state (section 11.2)

χ	ε_χ	$E(\chi)$	$E(\chi) - E_0$
2	2.5×10^{-1}	-3.89820374	3.6×10^{-1}
4	2.7×10^{-3}	-4.25110973	6.9×10^{-3}
8	4.6×10^{-6}	-4.25801999	1.5×10^{-5}
16	6.2×10^{-10}	-4.25803520	2.7×10^{-9}
32	0	-4.25803521	0

Heisenberg, $L = 10$: exact MPS, then truncated bond by bond from the left; ε_χ summed over the nine bonds (section 11.2).

- The error is *positive*: truncation can only raise the energy
- It is a few times ε_χ – here 1.5 to 4. **That is why ε_χ works as an error bar**
- Three to four decades per doubling of χ

Where lecture 1 leaves us

Established

- A bipartite state is a matrix; its SVD is the Schmidt decomposition
- Area law: $S \sim \ell^{\mathcal{D}-1}$ – constant in 1D, $\ln L$ if critical; generic states obey neither
- SVD at every bond $\Rightarrow$ MPS, exactly
- Truncation: best of its rank, variational, with a computable error

Missing

We built the MPS *from* a state vector – which we cannot have for large L .

- Tomorrow: working with an MPS without ever forming the vector
- Next week: *finding* the ground state by sweeping (sections 11.5–11.6)

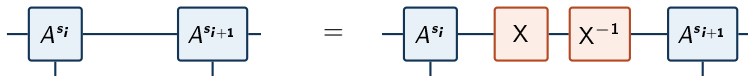
Exercises for this week: last frames – the stability matrix by Wick's theorem, and positive semidefinite matrices.

Friday, lecture 2: Gauge freedom, canonical forms and contractions

Gauge freedom (section 11.2)

$$A^{s_i} \rightarrow A^{s_i} X, \quad A^{s_{i+1}} \rightarrow X^{-1} A^{s_{i+1}}$$

leaves every coefficient unchanged: $\approx \chi^2$ redundant numbers per bond.



Canonical forms

$$\text{left: } \sum_s A^{s\dagger} A^s = I, \quad \text{right: } \sum_s B^s B^{s\dagger} = I,$$

$$\text{mixed: } C_{s_1 \dots s_L} = A^{s_1} \dots A^{s_{i-1}} \Theta^{s_i s_{i+1}} B^{s_{i+2}} \dots B^{s_L}$$

Lecture 1 already produced a left-canonical MPS: it is $U^\dagger U = I$, reshaped.

Why canonical forms matter (section 11.2)

$$|a_i\rangle_L = \sum_{s_1 \dots s_i} (A^{s_1} \dots A^{s_i})_{a_i} |s_1 \dots s_i\rangle \implies \langle a_i | a'_i \rangle_L = \delta_{a_i a'_i}$$

The block states are orthonormal – χ_i of them instead of d^i .

In mixed-canonical form

- Norm: $\langle \Psi | \Psi \rangle = \|\Theta\|_F^2$, no contraction of the chain
- The SVD of Θ is the Schmidt decomposition at that bond: entropy and truncation error **for free**
- Local expectation values are traces against Θ

Many-body connection

In CI the block basis is determinants: orthonormal, but $\binom{n}{N}$ of them. Here each half has a basis *adapted to the state* – χ block states chosen by the SVD. That is the sense in which DMRG is a renormalisation group.

Pause to think: gauge and canonical forms

Pause to think #4

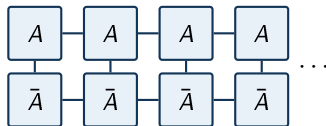
(about 4 min – discuss with your neighbour)

1. Gauge freedom removes $\approx \chi^2$ numbers per bond. How many parameters does the *state* have?
2. Left-canonical form can be reached by SVD or by QR. Why prefer QR?
3. Show that a left-canonical MPS is automatically normalised.

Answer to pause #4

1. $Ld\chi^2 - (L-1)\chi^2 \approx L\chi^2(d-1)$: the usual count overestimates by $d/(d-1)$. The redundancy buys trivial contractions.
2. QR is cheaper and *exact*: no singular values to inspect, nothing small to divide by.
QR to move the gauge, SVD only where you truncate.
3. Contract from the left: $\sum_{s_1} A^{s_1\dagger} A^{s_1} = I$, absorb into the next pair, repeat.
Canonicity does the normalisation – which is why the AKLT tensors needed none.

Contractions: the cost is in the order (section 11.2)



Transfer matrix $\mathbb{E} = \sum_s A^s \otimes \bar{A}^s$, applied one site at a time.

- Column by column: $\mathcal{O}(L\chi^3 d)$. An intermediate with three bond indices: χ^4
- Never form the d^L vector
- In mixed-canonical form a local $\hat{O}_i$ collapses to $\Theta, \hat{O}_i, \bar{\Theta}$: the rest cancels
- So $\langle \hat{n}_p \rangle$ and $\langle S_i \cdot S_{i+1} \rangle$ are free once the sweep passes

The most important practical habit in tensor-network computing.

Numerical check: the AKLT transfer matrix (section 11.2)

$\mathbb{E} = \sum_s \mathbf{A}^s \otimes \mathbf{A}^s$ is 4×4 , eigenvalues $\{1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}\}$.

- $\lambda_1 = 1$ *must* hold: canonicity as an eigenvalue equation
- $\xi = -1/\ln |\lambda_2| = 1/\ln 3$
- $\lambda_2 < 0$: alternating sign

r	$\langle S_2^z S_{2+r}^z \rangle$	$\frac{4}{3}(-\frac{1}{3})^r$
1	-0.444505	-0.444444
2	+0.148349	+0.148148
3	-0.049992	-0.049383
4	+0.018290	+0.016461

Open chain, $L = 10$: the deviation grows as the pair approaches the end.

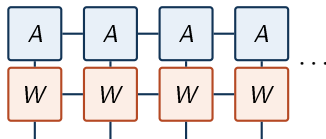
A finite χ *forces* exponential decay – so a critical chain is never an exact MPS.

Friday, lecture 2 continued: Matrix product operators and fermions

Matrix product operators (section 11.3)

$$\hat{H} = \sum_{\{s\}, \{s'\}} W^{s_1 s'_1} W^{s_2 s'_2} \dots W^{s_L s'_L} |s_1 \dots s_L\rangle \langle s'_1 \dots s'_L|$$

W_i is $D_{i-1} \times D_i$, and each *entry* is a $d \times d$ operator.



- $\hat{H}|\Psi\rangle$ is again an MPS, with bond dimension χD
- *Automaton*: a machine walks the chain, placing operators, remembering what is open
- D counts the partial terms open at a cut – not the terms in $\hat{H}$

The Heisenberg chain: $D = 5$ (section 11.3)

$$\hat{H} = J \sum_i S_i \cdot S_{i+1}, \quad W = \begin{pmatrix} \hat{1} & S^x & S^y & S^z & 0 \\ 0 & 0 & 0 & 0 & JS^x \\ 0 & 0 & 0 & 0 & JS^y \\ 0 & 0 & 0 & 0 & JS^z \\ 0 & 0 & 0 & 0 & \hat{1} \end{pmatrix}$$

First site: the first row. Last site: the last column.

- The five states: nothing placed; S^x , S^y or S^z waiting for a partner; finished
- Each path $1 \rightarrow \alpha \rightarrow 5$ places exactly one bond term
- Ising: $D = 3$. AKLT: $D = 14$, since the biquadratic term must remember which of *nine* pairs (α, β) is open, $1 + 3 + 9 + 1$

The pairing model: $D = 4$, though every level couples to every other (section 11.3)

$$\hat{H} = \sum_p \varepsilon_p \hat{n}_p - \frac{g}{2} \sum_{pq} \hat{P}_p^+ \hat{P}_q^-, \quad W_p = \begin{pmatrix} \hat{1} & \hat{P}_p^+ & \hat{P}_p^- & \varepsilon_p \hat{n}_p - \frac{g}{2} \hat{n}_{p+} \hat{n}_{p-} \\ 0 & \hat{1} & 0 & -\frac{g}{2} \hat{P}_p^- \\ 0 & 0 & \hat{1} & -\frac{g}{2} \hat{P}_p^+ \\ 0 & 0 & 0 & \hat{1} \end{pmatrix}$$

- $1 \rightarrow 4$: one-body and $p = q$; $1 \rightarrow 2 \rightarrow 4$: $q > p$; $1 \rightarrow 3 \rightarrow 4$: $q < p$
- The $\hat{1}$ on the diagonal *carries* an open term across any distance
- **Long range is free here**: the interaction factorises, v_{pq} has rank one
- A general two-body interaction does not – then D grows and ordering matters

Three more, and the systematic route (section 11.3)

Bond dimensions to know

- Hubbard chain: $D = 6$ (ring: 10)
- Penalty $\lambda(\hat{N} - N_0)^2$: $D = 3$
- Particle-hole term: $D = 14$
- Range- r hopping: $D = 2r + 2$

Do not write automata by hand for long

Write $\hat{H} = \sum_k c_k \hat{o}_1^{(k)} \cdots \hat{o}_L^{(k)}$, form at each cut the matrix of left parts against right parts, and compress it with an SVD: its rank *is* the minimal D .

A trap to avoid

Build the MPO and the exact-diagonalisation reference from the *same* operator-string list. Then the two solvers cannot disagree about a sign, and any discrepancy is a real bug.

Adding, applying, compressing (section 11.4)

Two exact operations that grow χ

- Sum: $N^s = M^s \oplus \tilde{M}^s$ (block diagonal), bond dimensions *add*
- MPO on MPS: $N_{(ba),(b'a')}^s = \sum_{s'} W_{bb'}^{ss'} M_{aa'}^{s'}$, bond dimension $D\chi$, cost $\mathcal{O}(Ld^2D^2\chi^2)$
- $\hat{H}^2$: an MPO of dimension $\leq D^2$, compressible (9, not 25, for Heisenberg)

One approximate operation that shrinks it

- SVD compression: sweep, truncate every bond; fast, one-sided errors
- Variational: minimise $\| |\psi\rangle - |\tilde{\psi}\rangle \|^2$ one tensor at a time; in mixed-canonical form the update is explicit, $\tilde{M} = \mathcal{L} M \mathcal{R}$, no linear system
- Seed the second with the first; two-site updates avoid getting stuck

MPS of fixed χ are not closed under $+$ or under $\hat{H}$; **compression is where every MPS algorithm makes its approximation.**

Pause to think: what the bond dimension counts

Pause to think #5

(about 4 min – discuss with your neighbour)

1. An $L = 100$ Heisenberg chain has 297 terms, and $D = 5$. Explain the discrepancy in one sentence.
2. What is D for $J_1 \sum_i S_i \cdot S_{i+1} + J_2 \sum_i S_i \cdot S_{i+2}$?
3. Pairing couples every level to every other and still has $D = 4$. What property does the work? Would a general $\sum_{pqrs} \langle pq | \hat{v} | rs \rangle a_p^\dagger a_q^\dagger a_s a_r$ keep it?

Answer to pause #5

1. D counts terms open *at one cut*, and at most one bond term straddles a cut; the 297 terms are *paths* through the network, not its width.
2. $D = 8$: each component now needs two waiting states, $1 + 3 + 3 + 1$.
3. **Factorisation**: $\sum_{pq} \hat{P}_p^+ \hat{P}_q^- = (\sum_p \hat{P}_p^+) (\sum_q \hat{P}_q^-)$, i.e. v_{pq} has rank one. A general interaction needs one open index per straddling pair, so $D = \mathcal{O}(L^2)$ for chemistry. Pairing is lucky, and we can name the reason.

Fermions on a chain: Jordan–Wigner (section 11.7; chapter 3, section 3.18)

An MPS describes *distinguishable* sites, so antisymmetry is put in by hand:

$$a_m^\dagger \longrightarrow (\text{local raising operator}) \times \prod_{m' < m} (-1)^{\hat{n}_{m'}}.$$

In practice: one site per level, $d = 4$, carrying $\hat{F} = (-1)^{\hat{n}_+ + \hat{n}_-}$ on every site to the left.

- Pairing: the strings **cancel** – $\hat{P}_p^\pm$ has both operators on the same site
- Hubbard and particle–hole: strings are real, but cost *no* bond dimension
- What they cost is **locality**: on a ring or in 2D, a local interaction becomes a string of length $\mathcal{O}(L)$

That is a large part of why the DMRG is a one-dimensional method.

Summary of week 40

Thursday: the ansatz

- A state is a matrix; its SVD is the Schmidt decomposition; U, V are isometries
- Area law: $S \sim \ell^{\mathcal{D}-1}$; constant only in $\mathcal{D} = 1$, $\ln L$ if critical
- SVD at every bond $\Rightarrow$ MPS,
 $\chi_i = \min(d^i, d^{L-i}, \chi)$
- $Ld\chi^2$ parameters; ε_χ is the error bar

Friday: the technology

- Canonical forms fix the gauge: orthonormal block states
- Mixed-canonical: norm = $\|\Theta\|_F^2$, entropies free
- Contract in the right order: $\mathcal{O}(L\chi^3 d)$
- MPO = automaton; D counts open terms: 5, 4, 3, 6, 14, $2r + 2$
- Jordan–Wigner: free in D , expensive in locality

Exercises: week 40, and answers for week 39

Exercise session: week 40 (suggested set)

Two exercises on the stability matrix of week 39, done with Wick's theorem and with linear algebra. Both are new and are answered on next week's slides.

Exercise 1: the stability matrix from Wick's theorem (section 6.10; new)

Write $\hat{H} = E_0 + \hat{F}_N + \hat{V}_N$, normal ordered with respect to the Hartree-Fock determinant $|c\rangle$, with $f_{pq} = \varepsilon_p \delta_{pq}$ in the Hartree-Fock basis.

- Show $\langle c'|c'\rangle = 1 + \sum_{ai} |\delta C_{ai}|^2 + \mathcal{O}(\delta C^3)$ for $|c'\rangle = e^{\hat{T}}|c\rangle$, saying which terms of $e^{\hat{T}} = \prod_i (1 + \hat{T}_i)$ are second order.
- Evaluate $\langle c|\{a_i^\dagger a_a\} \hat{F}_N \{a_b^\dagger a_j\}|c\rangle$ with Wick's theorem: draw the two complete contraction patterns, fix their signs, and show that the result is $f_{ab}\delta_{ij} - f_{ji}\delta_{ab} = (\varepsilon_a - \varepsilon_i)\delta_{ab}\delta_{ij} = \Delta_{ai,bj}$.
- Evaluate $\langle c|\{a_i^\dagger a_a\} \hat{V}_N \{a_b^\dagger a_j\}|c\rangle$: four complete contraction patterns, all equal by the antisymmetry of $\langle pq|\hat{v}|rs\rangle$, and the result is $A_{ai,bj} = \langle aj|\hat{v}|ib\rangle_{AS} = -\langle aj|\hat{v}|bi\rangle_{AS}$. Do the sign of one pattern in full.
- Evaluate $\langle c|\hat{V}_N a_a^\dagger a_i a_b^\dagger a_j|c\rangle$ and show that it equals $\langle ij|\hat{v}|ab\rangle_{AS} = B_{ai,bj}^*$, that E_0 and $\hat{F}_N$ contribute nothing to it, and that it vanishes unless $a \neq b$ and $i \neq j$.
- Assemble $\Delta E = \frac{1}{2}\langle\chi|\hat{M}|\chi\rangle$, verify that $\hat{M}$ is Hermitian, and check that for a single pair (a, i) it reduces to the curvature $\langle\Phi_i^a|\hat{H}|\Phi_i^a\rangle - E_0 = \varepsilon_a - \varepsilon_i - \langle ai|\hat{v}|ai\rangle$ of week 39, Thursday, with $B_{ai,ai} = \langle aa|\hat{v}|ii\rangle = 0$.

Exercise 2: the stability condition and positive semidefinite matrices (new)

Reminder

A Hermitian matrix $\hat{M}$ is *positive semidefinite*, $\hat{M} \geq 0$, if $\langle \chi | \hat{M} | \chi \rangle \geq 0$ for every χ . Equivalent statements: every eigenvalue is ≥ 0 ; $\hat{M} = C^\dagger C$ for some C ; every principal minor (the determinant of every principal submatrix, not only the leading ones) is ≥ 0 . **Necessary but not sufficient** conditions, each a consequence of the definition and cheap to check: every diagonal element $M_{kk} \geq 0$; every 2×2 principal minor $M_{kk}M_{ll} - |M_{kl}|^2 \geq 0$; $\text{tr } \hat{M} \geq 0$; $\det \hat{M} \geq 0$; every principal submatrix is itself ≥ 0 . Sylvester's criterion – all *leading* principal minors > 0 – characterises positive *definite* matrices; with ≥ 0 it is only necessary.

- Show that $\langle \chi | \hat{M} | \chi \rangle \geq 0$ for all χ is equivalent to all eigenvalues being non-negative, and that $\hat{M} = C^\dagger C$ implies it. Which of the two is the statement “ $\Delta E \geq 0$ in every direction of the manifold of determinants”?
- Show that $\text{diag}(0, -1)$ has all leading principal minors ≥ 0 and is not positive semidefinite. Explain why $\Delta + A \geq 0$, the necessary condition of week 39, is the “principal submatrix” condition, and why the diagonal condition $\varepsilon_a - \varepsilon_i + A_{ai, ai} \geq 0$ is weaker still.
- Lipkin model with $N = 2$, ε , V , $W = 0$: the two $1p$ - $1h$ pairs give a 4×4 matrix $\hat{M}$ with $\Delta + A = \varepsilon 1$ and $B = \begin{pmatrix} 0 & V \\ V & 0 \end{pmatrix}$. Write $\hat{M}$ out, check the conditions of the reminder one by one, and find the first one that fails as $|V|$ grows. Compare with the eigenvalues $\varepsilon \pm V$ and with $\chi = |V|(N-1)/\varepsilon$.
- Pairing model: a 2×2 block of $\hat{M}$ is $\begin{pmatrix} \Delta & -g/2 \\ -g/2 & \Delta \end{pmatrix}$ with $\Delta = \varepsilon_q - \varepsilon_p + g/2$. Check the 2×2 minor and find the eigenvalues. Why does the interaction cancel exactly in the lower one, and what does that say about breaking a pair coherently?
- Suppose $\Delta + A$ itself has a negative eigenvalue. What does the corresponding eigenvector mean physically, what would a CIS calculation on this reference return, and why is this case “worse” than an instability that appears only through B ?

Answers to the week-39 exercises: 1, varying one orbital

1a) Only components towards unoccupied orbitals matter

Insert $\phi_i + \sum_{\lambda} d_{\lambda} \phi_{\lambda}$ in row i and use linearity in that row: the component $d_j \phi_j$ with $j \leq F$, $j \neq i$, produces a determinant with two identical rows, which vanishes; $d_i \phi_i$ rescales Φ_0 by $1 + d_i$. What is left is $\Phi_0 + \sum_{a>F} d_a \Phi_i^a$, and since the Φ_i^a are orthonormal and orthogonal to Φ_0 , $N_i^{-2} = 1 + \sum_a |d_a|^2$.

1b) The expansion, and what a real η misses

$E(\eta) = [E_0 + \eta^* f_{ai} + \eta f_{ia} + |\eta|^2 H_{aa}] / (1 + |\eta|^2)$ with $H_{aa} = \langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle$. With $(1 + |\eta|^2)^{-1} = 1 - |\eta|^2 + \mathcal{O}(|\eta|^4)$,

$$E(\eta) = E_0 + 2 \operatorname{Re}(\eta f_{ia}) + |\eta|^2 (H_{aa} - E_0) - 2|\eta|^2 \operatorname{Re}(\eta f_{ia}) + \mathcal{O}(|\eta|^4),$$

the cubic term being the product of the linear term and $-|\eta|^2$. Stationarity needs $\operatorname{Re}(\eta f_{ia}) = 0$ for all η ; a real η fixes $\operatorname{Re} f_{ia} = 0$ and leaves $\operatorname{Im} f_{ia}$ free.

Answers to the week-39 exercises: 1, continued

1c) $\langle \Phi_0 | \hat{V}_N | \Phi_i^a \rangle = 0$ without Wick, and the block-diagonal Fock matrix

Write $\hat{V}_N$ in the particle-hole form of week 37: each term is a normal-ordered product of k quasi-particle creators and $4 - k$ quasi-particle annihilators (a hole annihilator $a_i^\dagger$ counts as a creator, and so on). $\langle \Phi_0 |$ is the quasi-particle vacuum, so any creator standing to the left kills the matrix element; only the term with four annihilators survives, and it needs four quasi-particles to annihilate. $|\Phi_i^a\rangle$ has two. Hence zero. – With $f_{ai} = 0$ for all pairs the Fock matrix in the basis (occupied first) is block diagonal, $f = \text{diag}(f_{oo}, f_{uu})$; diagonalising f_{oo} mixes occupied orbitals among themselves, which changes $|\Phi_0\rangle$ by a phase (section 6.3), and diagonalising f_{uu} does not touch it. Neither step disturbs $f_{ai} = 0$.

1d) The plot

For the random Hamiltonian the picture is the one of week 39's numerical-check frame: in the original basis $E(\eta) - E_0$ is a straight line of slope $2f_{ia}$ through the origin, bent by the quadratic term; its odd part equals $2\eta f_{ia}$ to within the cubic term $-2\eta^3 f_{ia}$; a purely imaginary η gives no linear change because f_{ia} is real. In the Hartree-Fock basis the slope is zero to the SCF tolerance and $E(\eta) - E_0 = \eta^2 (H_{aa} - E_0) + \mathcal{O}(\eta^4)$, a parabola whose curvature is the diagonal element of the stability matrix for that pair. (Trap system, for comparison: slope -0.867 at $(a, i) = (4, 2)$ in the oscillator basis, -1.7×10^{-10} in the Hartree-Fock basis.)

Answers to the week-39 exercises: 2, single-particle energies and Thouless

2a) $E^{\text{HF}} \neq \sum_i \varepsilon_i$

Multiply the Hartree-Fock equation $\sum_{\beta} h_{\alpha\beta}^{\text{HF}} C_{i\beta} = \varepsilon_i C_{i\alpha}$ by $C_{i\alpha}^*$, sum over α and $i \leq F$:

$\sum_i \varepsilon_i = \sum_i \langle i | \hat{h}_0 | i \rangle + \sum_{ij} \langle ij | \hat{v} | ij \rangle_{\text{AS}}$. The energy functional carries $\frac{1}{2}$ on the interaction, so $E^{\text{HF}} = \sum_i \varepsilon_i - \frac{1}{2} \sum_{ij} \langle ij | \hat{v} | ij \rangle_{\text{AS}}$: the interaction is counted once per partner in the sum of eigenvalues. In the notebook, $\sum_i \varepsilon_i - \frac{1}{2} \sum_{ij} \langle ij | \hat{v} | ij \rangle$ agrees with E^{HF} to 10^{-12} for the trap system.

2b) Koopmans

Removing orbital k with the others frozen: $E(N) - E(N-1) = \langle k | \hat{h}_0 | k \rangle + \sum_{j \neq k} \langle kj | \hat{v} | kj \rangle_{\text{AS}} = \varepsilon_k$ (the $j = k$ term vanishes by antisymmetry). Adding a particle in a gives $E(N+1) - E(N) = \varepsilon_a$ by the same algebra, but ε_a was computed in the field of *all* N particles, whereas the added particle should feel the other N and the occupied orbitals should relax around it; virtual orbitals are optimised for nothing, and the relaxation error is larger than for removal, where the orbital was at least self-consistent.

Answers to the week-39 exercises: 2, continued

2c) $[a_a^\dagger a_i, a_b^\dagger a_j] = 0$, $\hat{T}_i^2 = 0$, and the cross terms

With $i \neq b$: $a_i a_b^\dagger = -a_b^\dagger a_i$, so $a_a^\dagger a_i a_b^\dagger a_j = -a_a^\dagger a_b^\dagger a_i a_j$. With $j \neq a$: $a_b^\dagger a_j a_a^\dagger a_i = -a_b^\dagger a_a^\dagger a_j a_i = -a_a^\dagger a_b^\dagger a_j a_i$, the last step swapping both pairs (two signs). Equal, so the commutator vanishes; disjointness of the index sets is what prevented a δ from appearing in either anticommutation.

$\hat{T}_i^2 = \sum_{ab} C_{ai} C_{bi} a_a^\dagger a_i a_b^\dagger a_i = -\sum_{ab} C_{ai} C_{bi} a_a^\dagger a_b^\dagger a_i a_i = 0$, hence $e^{\hat{T}_i} = 1 + \hat{T}_i$ and $e^{\hat{T}} = \prod_i (1 + \hat{T}_i)$. The cross term $\hat{T}_i \hat{T}_j = \sum_{ab} C_{ai} C_{bj} a_a^\dagger a_b^\dagger a_j a_i$ ($i \neq j$) is a $2p-2h$ excitation with amplitude $C_{ai} C_{bj} - C_{bi} C_{aj}$ after antisymmetrisation: products of singles, no freedom of their own.

2d) $\langle c | \tilde{c} \rangle = \det(f_{ip})$

$N = 2$: $\langle 0 | a_2 a_1 (\sum_p f_{1p} a_p^\dagger) (\sum_q f_{2q} a_q^\dagger) | 0 \rangle$; only $p, q \in \{1, 2\}$ survive and $\langle 0 | a_2 a_1 a_p^\dagger a_q^\dagger | 0 \rangle = \delta_{p1} \delta_{q2} - \delta_{p2} \delta_{q1}$, giving $f_{11} f_{22} - f_{12} f_{21}$. In general every $a_p^\dagger$ must carry an occupied index and be contracted with a distinct a_{i_k} ; summing over assignments with their signs is $\sum_\pi \text{sgn}(\pi) \prod_k f_{i_k \pi(i_k)} = \det(f)$ over the occupied block. The proof inverts that block, so $\det \neq 0$ is exactly what it needs; if it vanishes, $|\tilde{c}\rangle$ is orthogonal to $|c\rangle$ and not in its neighbourhood at all.

Answers to the week-39 exercises: 3, stability

3a) The norm

$|c'\rangle = \{1 + \sum \delta C_{ai} a_a^\dagger a_i + \frac{1}{2} \sum \delta C_{ai} \delta C_{bj} a_a^\dagger a_i a_b^\dagger a_j + \dots\} |c\rangle$. Zeroth order: 1. Constant times linear: $\langle c | a_a^\dagger a_i | c \rangle = 0$. Linear times linear: $\sum \delta C_{ai}^* \delta C_{bj} \langle \Phi_i^a | \Phi_j^b \rangle = \sum |\delta C_{ai}|^2$. Constant times quadratic: $\langle c | a_a^\dagger a_i a_b^\dagger a_j | c \rangle = 0$. Linear times quadratic is third order, quadratic times quadratic fourth. Hence $1 + \sum |\delta C|^2 + \mathcal{O}(\delta C^3)$.

3b) ΔE , the doubled vector, Hermiticity, the necessary condition

Subtract $(1 + \sum |\delta C|^2) E_0$ from the second-order energy and divide by the norm: what remains over the norm is ΔE . The $\delta C^* \delta C$ terms are $\delta C^\dagger (\Delta + A) \delta C$, the $\delta C^* \delta C^*$ terms $\frac{1}{2} \delta C^\dagger B \delta C^*$, the $\delta C \delta C$ terms $\frac{1}{2} \delta C^T B^* \delta C$; stacking $\chi = (\delta C, \delta C^*)$ writes all three as $\frac{1}{2} \langle \chi | \hat{M} | \chi \rangle$. Hermitian: $(\Delta + A)_{bj, ai} = \langle bi | \hat{v} | ja \rangle = \langle aj | \hat{v} | ib \rangle^* = (\Delta + A)_{ai, bj}^*$ and $B_{bj, ai} = \langle ba | \hat{v} | ji \rangle = B_{ai, bj}$, so the lower-left block is $B^* = B^\dagger$ and $\hat{M}^\dagger = \hat{M}$. Restricting χ to $(\delta C, 0)$ – allowed, since $\langle \chi | \hat{M} | \chi \rangle \geq 0$ must hold for every χ – gives $\delta C^\dagger (\Delta + A) \delta C \geq 0$: necessary; not sufficient, because a general χ also feels B .

Answers to the week-39 exercises: 3, continued

3c) Lipkin, $W = 0$

$\frac{V}{2}(\hat{J}_+^2 + \hat{J}_-^2)$ changes J_z by ± 2 , so it has no element inside the $1p-1h$ space: $A = 0$, $\Delta = \varepsilon_1$. $\hat{J}_-^2$ connects the reference to the $2p-2h$ state with pairs p and q both raised, so $B_{p,q} = V$ for $p \neq q$ and 0 for $p = q$:

$B = V(J - 1)$, eigenvalues $V(N - 1)$ once and $-V(N - 1)$ times. With $\hat{M} = \begin{pmatrix} \varepsilon_1 & B \\ B & \varepsilon_1 \end{pmatrix}$ the eigenvalues are $\varepsilon \pm b$, and the smallest is $\varepsilon - |V|(N - 1) = \varepsilon(1 - \chi)$.

3d) Pairing, four levels, four particles: $\lambda_{\min}(\hat{M}) = 1$ for every g

The pairing interaction moves pairs, never a single particle: $A = 0$. Breaking the pair at p to put one particle in q costs $\varepsilon_q - \varepsilon_p$ plus the lost pairing energy $g/2$: $\Delta = \varepsilon_q - \varepsilon_p + g/2$, and $\lambda_{\min}(\Delta + A) = 1 + g/2 > 0$. B connects $\delta C_{q\uparrow,p\uparrow}$ with $\delta C_{q\downarrow,p\downarrow}^*$ through $\langle q\uparrow q\downarrow | \hat{v} | p\uparrow p\downarrow \rangle = -g/2$, so $\hat{M}$ splits into 2×2 blocks $\begin{pmatrix} \Delta & -g/2 \\ -g/2 & \Delta \end{pmatrix}$ with eigenvalues $\Delta \mp g/2 = \varepsilon_q - \varepsilon_p$ and $\varepsilon_q - \varepsilon_p + g$. Moving the two members of a pair *together* costs exactly the level spacing; the interaction cancels; the minimum is $\varepsilon_3 - \varepsilon_2 = 1$ for all g (notebook: 1.0000 from $g = 0.25$ to 3). Stable, and useless: no number-conserving mean field captures pairing, which is why chapter 7 breaks particle number (BCS) and why the DMRG of this week, which keeps it, has to carry the correlation in its bond dimension instead.

Next week, and a closing remark

Week 41: lectures 3 and 4

Thursday: the variational principle on the MPS manifold; environments and the effective Hamiltonian; the two-site update, Lanczos and SVD; extrapolation in ε_χ .

Friday: the pairing model, where the DMRG *discovers* seniority zero; the Hubbard chain past exact diagonalisation; TEBD, where Trotter–Suzuki returns; the limits in 2D and in chemistry.

Reading: chapter 11, sections 11.5–11.6 (the sweep, and White’s original blocks and superblocks), 11.8–11.12; Schollwöck, *Annals of Physics* 326, 96 (2011), sections 2 and 6.

One idea did all the work: the singular value decomposition. Applied once it is the Schmidt decomposition; applied at every bond it turns d^L amplitudes into L small matrices, with the optimal truncation and a computable error.

Next week we stop *building* matrix product states, and start *finding* them.